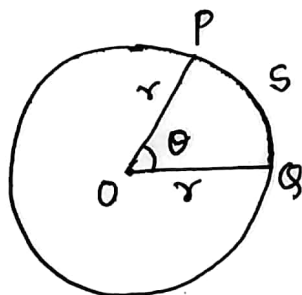


Trigonometry Formula Sheet

- $1^{\circ} = 60'$
- $1' = 60''$
- $1^{\circ} = 3600''$
- The sum of measures of exterior angles of a polygon is 360° or $2\pi^{\circ}$.



- length of arc $PQ = s = r\theta$ (θ is in radians)
- Area of sector $POQ = \frac{\theta}{360} \times \pi r^2$

Formula Sheet

Relation between degree measure and radian measure

- To convert degree measure into radian measure, multiply degree measure by $\frac{\pi}{180}$
- To convert radian measure into degree measure, multiply radian measure by $\frac{180}{\pi}$

Pythagorean Identities

- $\sin^2 x + \cos^2 x = 1$
- $\sec^2 x - \tan^2 x = 1$
- $\operatorname{cosec}^2 x - \cot^2 x = 1$

Reciprocal Identities

- $\sin x = \frac{1}{\operatorname{cosec} x}$
- $\operatorname{cosec} x = \frac{1}{\sin x}$
- $\cos x = \frac{1}{\sec x}$
- $\sec x = \frac{1}{\cos x}$
- $\tan x = \frac{1}{\cot x}$
- $\cot x = \frac{1}{\tan x}$

Odd-even Identities

- $\sin(-x) = -\sin x$
- $\cos(-x) = \cos x$
- $\tan(-x) = -\tan x$
- $\operatorname{cosec}(-x) = -\operatorname{cosec} x$
- $\sec(-x) = \sec x$
- $\cot(-x) = -\cot x$

Sum and Difference Formulae

- $\sin(A + B) = \sin A \cos B + \cos A \sin B$
- $\sin(A - B) = \sin A \cos B - \cos A \sin B$
- $\cos(A + B) = \cos A \cos B - \sin A \sin B$
- $\cos(A - B) = \cos A \cos B + \sin A \sin B$
- $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$
- $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$

Multiple angle Formulae

$$\checkmark \sin 2A = 2 \sin A \cos A$$

$$\checkmark \cos 2A = \cos^2 A - \sin^2 A$$

$$\checkmark \cos 2A = 2 \cos^2 A - 1$$

$$\checkmark \cos 2A = 1 - 2 \sin^2 A$$

$$\bullet \tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$\bullet \sin 2A = \frac{2 \tan A}{1 + \tan^2 A}$$

$$\bullet \cos 2A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$$

$$\bullet \sin 3A = 3 \sin A - 4 \sin^3 A$$

$$\bullet \cos 3A = 4 \cos^3 A - 3 \cos A$$

$$\bullet \tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}$$

All the formulae for 2θ can be modified for angle $\theta/2$

Power reducing formulae

$$\bullet \sin^2 A = \frac{1 - \cos 2A}{2}$$

$$\bullet \cos^2 A = \frac{1 + \cos 2A}{2}$$

$$\bullet \tan^2 A = \frac{1 - \cos 2A}{1 + \cos 2A}$$

$$1 - \cos 2A = 2 \sin^2 A$$

$$1 + \cos 2A = 2 \cos^2 A$$

Product to sum formulae

$$\bullet \sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

$$\bullet \cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$

$$\bullet \sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)]$$

$$\bullet \cos A \sin B = \frac{1}{2} [\sin(A + B) - \sin(A - B)]$$

Sum to Product formulae

$$\bullet \sin A + \sin B = 2 \sin \left(\frac{A + B}{2} \right) \cos \left(\frac{A - B}{2} \right)$$

$$\bullet \sin A - \sin B = 2 \cos \left(\frac{A + B}{2} \right) \sin \left(\frac{A - B}{2} \right)$$

$$\bullet \cos A + \cos B = 2 \cos \left(\frac{A + B}{2} \right) \cos \left(\frac{A - B}{2} \right)$$

$$\bullet \cos A - \cos B = -2 \sin \left(\frac{A + B}{2} \right) \sin \left(\frac{A - B}{2} \right)$$

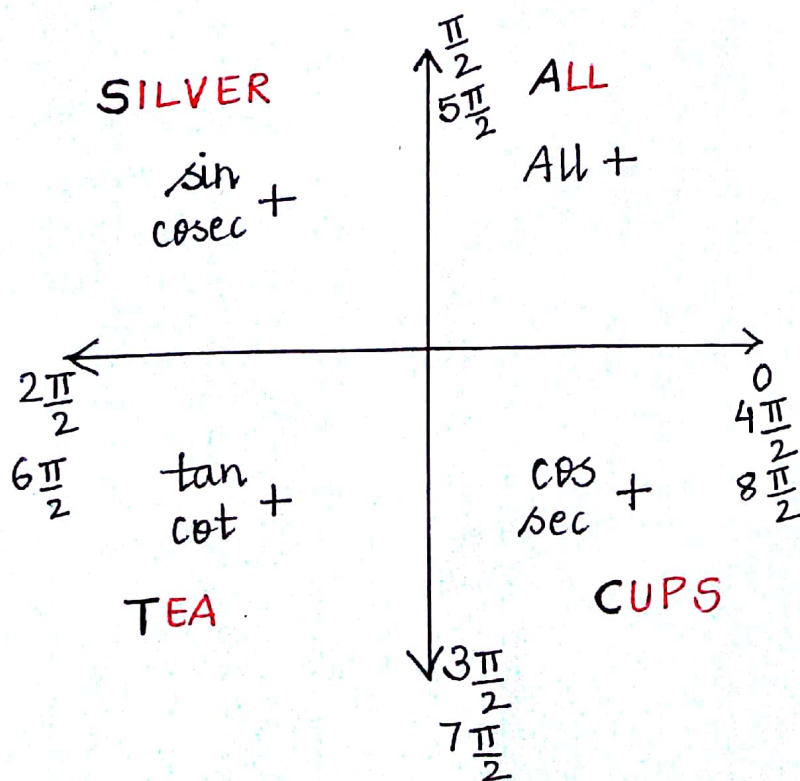
Allied Angles

- The sum or difference of two angles, one of which is an integral multiple of 90° or $\frac{\pi}{2}^c$ is said to be an allied angle.
- If θ is the measure of a given angle and $n \in \mathbb{I}$ then $n\frac{\pi}{2} \pm \theta$ are general forms of allied angles.

Rules for conversion

- 1) For $n\frac{\pi}{2} \pm \theta$, if n is odd then
- $$\begin{aligned} \sin &\longleftrightarrow \cos \\ \tan &\longleftrightarrow \cot \\ \sec &\longleftrightarrow \csc \end{aligned}$$
- if n is even then the given ratio remains same.

- 2) Sign of the converted term depends on the quadrant in which $n\frac{\pi}{2} \pm \theta$ lies.



Examples:

i) $\csc\left(\frac{3\pi}{2} - \theta\right) = -\sec \theta$

ii) $\sin(2\pi - \theta) = \sin\left(4\frac{\pi}{2} - \theta\right) = -\sin \theta$

iii) $\begin{aligned} \sec 510^\circ &= \sec(450^\circ + 60^\circ) \\ &= \sec\left(5\frac{\pi}{2} + 60^\circ\right) \\ &= -\csc 60^\circ \\ &= -\frac{2}{\sqrt{3}} \end{aligned}$

Trigonometric functions of $(\frac{\pi}{2} - \theta)$ and $(\frac{\pi}{2} + \theta)$

- $\sin(\frac{\pi}{2} - \theta) = \cos \theta$
- $\cos(\frac{\pi}{2} - \theta) = \sin \theta$
- $\tan(\frac{\pi}{2} - \theta) = \cot \theta$
- $\cot(\frac{\pi}{2} - \theta) = \tan \theta$
- $\operatorname{cosec}(\frac{\pi}{2} - \theta) = \sec \theta$
- $\sec(\frac{\pi}{2} - \theta) = \operatorname{cosec} \theta$
- $\sin(\frac{\pi}{2} + \theta) = \cos \theta$
- $\cos(\frac{\pi}{2} + \theta) = -\sin \theta$
- $\tan(\frac{\pi}{2} + \theta) = -\tan \theta$
- $\cot(\frac{\pi}{2} + \theta) = -\cot \theta$
- $\operatorname{cosec}(\frac{\pi}{2} + \theta) = \sec \theta$
- $\sec(\frac{\pi}{2} + \theta) = -\operatorname{cosec} \theta$

Trigonometric functions of $(\pi - \theta)$ and $(\pi + \theta)$

- $\sin(\pi - \theta) = \sin \theta$
- $\cos(\pi - \theta) = -\cos \theta$
- $\tan(\pi - \theta) = -\tan \theta$
- $\cot(\pi - \theta) = -\cot \theta$
- $\operatorname{cosec}(\pi - \theta) = \operatorname{cosec} \theta$
- $\sec(\pi - \theta) = -\sec \theta$
- $\sin(\pi + \theta) = -\sin \theta$
- $\cos(\pi + \theta) = -\cos \theta$
- $\tan(\pi + \theta) = \tan \theta$
- $\cot(\pi + \theta) = \cot \theta$
- $\operatorname{cosec}(\pi + \theta) = -\operatorname{cosec} \theta$
- $\sec(\pi + \theta) = -\sec \theta$

Trigonometric functions of $(\frac{3\pi}{2} - \theta)$ and $(\frac{3\pi}{2} + \theta)$

- $\sin(\frac{3\pi}{2} - \theta) = -\cos \theta$
- $\cos(\frac{3\pi}{2} - \theta) = -\sin \theta$
- $\tan(\frac{3\pi}{2} - \theta) = \cot \theta$
- $\cot(\frac{3\pi}{2} - \theta) = \tan \theta$
- $\operatorname{cosec}(\frac{3\pi}{2} - \theta) = -\sec \theta$
- $\sec(\frac{3\pi}{2} - \theta) = -\operatorname{cosec} \theta$
- $\sin(\frac{3\pi}{2} + \theta) = -\cos \theta$
- $\cos(\frac{3\pi}{2} + \theta) = \sin \theta$
- $\tan(\frac{3\pi}{2} + \theta) = -\tan \theta$
- $\cot(\frac{3\pi}{2} + \theta) = -\cot \theta$
- $\operatorname{cosec}(\frac{3\pi}{2} + \theta) = -\sec \theta$
- $\sec(\frac{3\pi}{2} + \theta) = \operatorname{cosec} \theta$

Trigonometric functions of $(2\pi - \theta)$ and $(2\pi + \theta)$

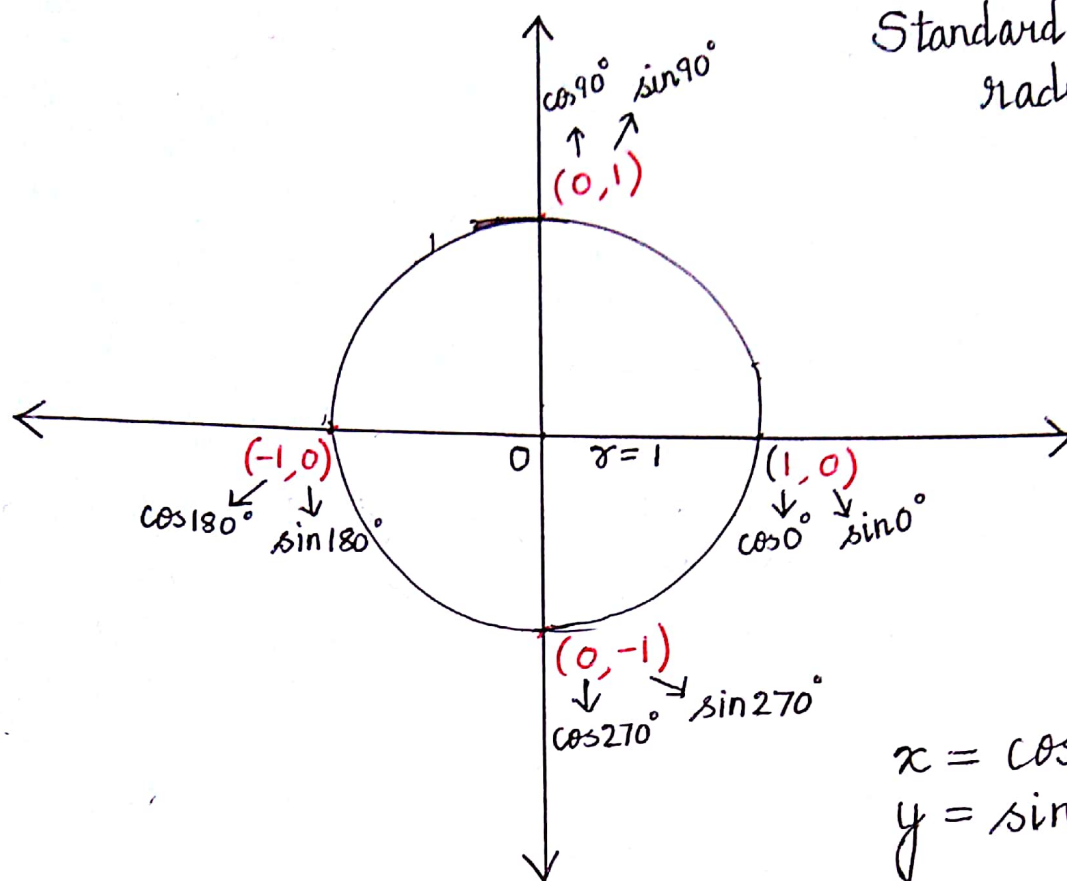
- $\sin(2\pi - \theta) = -\sin \theta$
- $\cos(2\pi - \theta) = \cos \theta$
- $\tan(2\pi - \theta) = -\tan \theta$
- $\cot(2\pi - \theta) = -\cot \theta$
- $\operatorname{cosec}(2\pi - \theta) = -\operatorname{cosec} \theta$
- $\sec(2\pi - \theta) = \sec \theta$
- $\sin(2\pi + \theta) = \sin \theta$
- $\cos(2\pi + \theta) = \cos \theta$
- $\tan(2\pi + \theta) = \tan \theta$
- $\cot(2\pi + \theta) = \cot \theta$
- $\operatorname{cosec}(2\pi + \theta) = \operatorname{cosec} \theta$
- $\sec(2\pi + \theta) = \sec \theta$

Properties of Inverse Trigonometric functions

- $\sin^{-1}\left(\frac{1}{x}\right) = \operatorname{cosec}^{-1} x$
- $\cos^{-1}\left(\frac{1}{x}\right) = \sec^{-1} x$
- $\tan^{-1}\left(\frac{1}{x}\right) = \cot^{-1} x$
- $\sin^{-1}(-x) = -\sin^{-1}(x)$
- $\tan^{-1}(-x) = -\tan^{-1}(x)$
- $\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1}(x)$
- $\cos^{-1}(-x) = \pi - \cos^{-1}(x)$
- $\sec^{-1}(-x) = \pi - \sec^{-1}(x)$
- $\cot^{-1}(-x) = \pi - \cot^{-1}(x)$
- $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$
- $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$
- $\operatorname{cosec}^{-1} x + \sec^{-1} x = \frac{\pi}{2}$
- $2 \tan^{-1} x = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$
- $2 \tan^{-1} x = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$
- $2 \tan^{-1} x = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$
- $\tan^{-1} x + \tan^{-1} y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$, if $x, y > 0$ and $xy < 1$
- $\tan^{-1} x + \tan^{-1} y = \pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right)$, if $x, y > 0$ and $xy > 1$
- $\tan^{-1} x - \tan^{-1} y = \tan^{-1}\left(\frac{x-y}{1+xy}\right)$, where $x > 0, y > 0$

Sine rule, cosine rule and projection rule

- In $\triangle ABC$, $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$ (sine rule)
- In $\triangle ABC$,
 1. $a^2 = b^2 + c^2 - 2bc \cos A$
 2. $b^2 = c^2 + a^2 - 2ac \cos B$
 3. $c^2 = a^2 + b^2 - 2ab \cos C$ (cosine rule)
- In $\triangle ABC$,
 1. $a = c \cos B + b \cos C$
 2. $b = a \cos C + c \cos A$
 3. $c = a \cos B + b \cos A$ (projection rule)



$$x = \cos \theta$$

$$y = \sin \theta$$

Angle (degrees/radians)								
Trigonometric functions	0°	30°	45°	60°	90°	180°	270°	360°
	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$	π	$3\pi/2$	2π
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	0	-1	0
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	-1	0	1
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞	0	∞	0
$\operatorname{cosec} \theta$	∞	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1	∞	-1	∞
$\sec \theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	∞	-1	∞	1
$\cot \theta$	∞	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0	∞	0	∞

Function	Domain	Range
$y = \sin (x)$	All Real Numbers	$\{y \mid -1 \leq y \leq 1\}$
$y = \cos (x)$	All Real Numbers	$\{y \mid -1 \leq y \leq 1\}$
$y = \csc (x)$	$\{x \mid x \neq \dots\dots\dots -2\pi, -\pi, 0, \pi, 2\pi\dots\dots\dots\}$	$\{y \mid y \leq -1 \text{ or } y \geq 1\}$
$y = \sec (x)$	$\{x \mid x \neq \dots\dots\dots -3\pi/2, -\pi/2, \pi/2, 3\pi/2\dots\dots\dots\}$	$\{y \mid y \leq -1 \text{ or } y \geq 1\}$
$y = \tan (x)$	$\{x \mid x \neq \dots\dots\dots -3\pi/2, -\pi/2, \pi/2, 3\pi/2\dots\dots\dots\}$	All Real Numbers
$y = \cot (x)$	$\{x \mid x \neq \dots\dots\dots -2\pi, -\pi, 0, \pi, 2\pi\dots\dots\dots\}$	All Real Numbers

Function	Domain	Range
$y = \sin^{-1}(x)$	$\{ x \mid -1 \leq x \leq 1 \}$	$\{ y \mid -\pi/2 \leq y \leq \pi/2 \}$
$y = \cos^{-1}(x)$	$\{ x \mid -1 \leq x \leq 1 \}$	$\{ y \mid 0 \leq y \leq \pi \}$
$y = \csc^{-1}(x)$	$\{ x \mid x \leq -1 \text{ or } x \geq 1 \}$	$\{ y \mid -\pi/2 \leq y \leq \pi/2, y \neq 0 \}$
$y = \sec^{-1}(x)$	$\{ x \mid x \leq -1 \text{ or } x \geq 1 \}$	$\{ y \mid 0 \leq y \leq \pi, y \neq \pi/2 \}$
$y = \tan^{-1}(x)$	All Real Numbers	$\{ y \mid -\pi/2 < y < \pi/2 \}$
$y = \cot^{-1}(x)$	All Real Numbers	$\{ y \mid 0 < y < \pi \}$